Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Faculty of Engineering Sciences (FES)

MT-101 (Section H, CS & AI) Calculus I

Pre-Requisite(s): None

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Office: G-14, Faculty of Basic Sciences

Office hours: 2.30pm to 30pm

Course Introduction

This is an introductory course of 'Calculus' required for all engineering students. The pre-requisite is the mathematics taught at intermediate / A level to students in Pakistan. There will be quick review of the Calculus studied by the students in their intermediate (F.Sc. / A level) classes but at a much-advanced level, with introduction of many new topics and material. The emphasis will be on the application of 'Differential and Integral Calculus' to problems of physical sciences and engineering. At the end of the course, the students should be able to tackle the problems in other disciplines that require calculus tools for their solution.

Course Contents

- **Functions:** Real numbers; Functions and their graphs; Basic elementary functions; Combining functions; Elementary functions; Domain and range of functions; Shifting and scaling of graphs.
- **Limit and Continuity:** Limit of a function; Calculation of limits and limit laws; Limits involving infinity; One-sided limits; Continuous and discontinuous functions; Types of discontinuity; Asymptotes of graphs.
- **Differential Calculus I:** Rate of change and tangents to curves; Derivative at a point; Geometric interpretation; Differentiation rules; Derivative as rate of change; Derivatives of basic elementary functions; The chain rule; Implicit differentiation; Related rates; Linearization and differentials.
- **Differential Calculus II:** Derivatives of transcendental functions; Inverse functions and their derivatives; Exponential and logarithmic functions; Trigonometric and inverse trigonometric functions; Hyperbolic functions; Logarithmic differentiation; Intermediate forms and L'Hopital's rule.

- **Applications of Derivatives:** Mean value theorem; Monotonic functions and the first derivative test; Extreme values of functions; Concavity and the second derivative test; Applied optimization.
- **Integral Calculus:** Anti-derivatives and indefinite integrals; Elementary integration techniques; Sigma notation and limits of finite sums; The definite integral; The fundamental theorem of calculus; Elementary properties and calculation of definite integrals.
- **Applications of Integral Calculus:** Area under a curve and between curves; Volumes using cross-sections; Volumes using cylindrical shells; Arc length; Areas of surfaces of revolution; Calculation of work; Center of gravity of plane laminas.
- **Techniques of Integration:** Use of basic integration formulas; integration by substitution and by parts; Trigonometric integrals; Trigonometric substitutions; Integration of rational functions by partial fractions; Numerical integration; Improper integrals.
- **Infinite Series:** Definitions, convergence, properties. Integral test, basic and limit comparison tests, ratio and root tests. Alternating series and absolute convergence. Power series. McLaurin and Taylor series. Applications of power series.

Mapping of Class Learning Outcome (CLOs) to Program Learning Outcomes (PLOs)

S. No	CLOs	PLOs	Bloom Taxonomy
Upon completion of this course, students will be able to:			
CLO-1	Solve problems related to limit and continuity of a function and their inter-relationship.	PLO-1	C3 (Applying)
CLO-2	Calculate the derivative and differential of a function and apply them in different applied problems.	PLO-1	C3 (Applying)
CLO-3	Use different techniques of integration to solve different applied problems.	PLO-1	C3 (Applying)
CLO-4	Apply different tests to discuss the convergence of sequences and series.	PLO-1	C3 (Applying)

Grading Policy

Assessment Items		% Marks
1.	**Announced Quizzes**	**20%**
3.	**Assignments**	**10%**
4.	**Mid-Term Exam**	**30%**
5.	**Final Exam**	**40%**

Text and Reference Books

Textbooks:

- **"Thomas' Calculus"** by George B. Thomas, Jr., Maurice D. Weir, Joel R. Hass. Christopher Heil, 13th Edition 2010. Pearson, USA.

Reference books:

- **"Calculus: Early Transcendentals"** by James Stewart. 6th Edition 2008. Brooks/Cole USA.
- **"Calculus"** by Swokowski, Olinick, Pence. 6th Edition 1994. PWS, USA.

Recommended online resources:

- https://www.symbolab.com. https://www.geogebra.org. https://www.desmos.com.

Course Policy

Classes will consist of lectures in the form of slides and/or notes written on the board as well as sample problems/examples. Students should take written notes during class of any material the instructor covers on the board. Lectures and class slides are **not exhaustive**, and students should use them in conjunction with the pre-assigned textbook readings, notes/examples taken during lectures and assigned homework problems to learn the course topics. The problems in assignments (homework) are mostly taken from the textbook and will be emailed to the students. The purpose of the assignments is to assist students in learning about the course topics and to prepare them for the quizzes and exams, which will contain questions similar to those in the assigned homework problems. Students may work on home assignments in collaboration with each other, but they must submit their own work; no copying from others. All the direct assessment tools, i.e., quizzes, assignment, midterms, project and final exams, except surprised quizzes, will be announced (by email) well before time. All quizzes/examinations will be closed book. Calculators and mobile phones are strictly prohibited in exams and quizzes. Students **MUST** attempt/submit all the exams/assignments. Failure to attempt/submit any of the assessment may result in failure in that assessment. Late submission of assignments (homework) and retaking of any quiz/exam will **NOT** be accepted/allowed. Student Attendance is expected to be 100%, and a minimum of 80% (mandatory) attendance is required to take the final exams. Students are welcome to visit the instructor during the office time for any queries or concerns.

Academic Honesty Policy

FES follows a strict Academic Honesty Policy that applies to this course as well. Any violations can lead to deduction of marks, referral to faculty Academic Honesty Committee, issuing of warning letter (with copy to parents), referral to Institute Disciplinary Committee, award of F grade in course or entire semester, and possible removal from rolls of the institute. Students must read and adhere to the policy available on webpage and with Student Office

Detailed Course Contents	
Topic	**Book Section**
Real Numbers and the Real Line.	**(Self-Study)** Appendix A.1
Functions and their Graphs	1.1
Trigonometric Functions, Exponential Functions	**(Self-Study)** 1.3, 1.5
Combining Functions; Shifting and Scaling Graphs.	1.2
Inverse Functions and Logarithms	1.6
Rates of Change and Tangents to Curves	**(Self-Study)** 2.1
Limit of a Function and limit laws	2.2
One-Sided limits	2.4
Continuity	2.5
Limits involving infinity; Asymptotes of graphs:	2.6
Tangents and the Derivative at a Point	3.1
The Derivative as a Function	3.2
Differentiation Rules	3.3
Derivatives of Trigonometric Functions	3.5
The Chain Rule	3.6
Implicit differentiation	3.7
Derivatives of Inverse Functions and Logarithms	3.8
Inverse Trigonometric Functions	3.9
Extreme Values of Functions	4.1
The Mean Value Theorem	4.2
Monotonic Functions and the First Derivative Test	4.3
Concavity and Curve Sketching	4.4
Indeterminate Forms and L'Hôpital's Rule	4.5
Applied Optimization	4.6
Antiderivatives	4.8
Area and Estimating with Finite Sums	5.1
Sigma Notation and Limits of Finite Sums	5.2
The Definite Integral	5.3
The Fundamental Theorem of Calculus	5.4
Indefinite Integrals and the Substitution Method	5.5
Definite Integral Substitutions and the Area Between Curves	5.6
Volumes Using Cross-Sections	6.1
Volumes Using Cylindrical Shells	6.2
Arc Length	6.3
Areas of Surfaces of Revolution	6.4
Work and Fluid Forces	6.5
Moments and Centers of Mass	6.6
Using Basic Integration Formulas	8.1
Integration by Parts	8.2
Trigonometric Integrals	8.3

Trigonometric Substitutions	8.4
Integration of Rational Functions by Partial Fractions	8.5
Improper Integrals	8.8
Sequences	10.1
Infinite Series	10.2
The Integral Test	10.3
Comparison Tests	10.4
Absolute Convergence; The Ratio and Root Tests	10.5
Alternating Series and Conditional Convergence	10.6
Power Series	10.7
Taylor and Maclaurin Series	10.8
Convergence of Taylor Series	10.9
Quick review of functions of 2 & 3 variables and their partial derivatives	14.1
Quick review of functions of 2 & 3 variables and their partial derivatives. Continue	14.3

Note: This outline serves only as rough guidance for the course. It may be changed or modified as and when deemed necessary by the instructor. The instructor is at liberty to best distribute the number of lectures and/or change the sequence of topics to cover the entire course.